

2-meter quiz for today, but I'll do that after the start of the next lecture hour, I think, than more people are here.

And this will be about the midterm test. I hope you will survive that. We'll discuss about that in the 2-meter quiz. Today I'm going to wrap up module 6 on light, on light propagation. And today we're going to look at polarization, scattering, interference, and diffraction. But first, let's see if you remember what I covered last Thursday. It wasn't too crowded then because people were preparing for the midterm test, but let's see if you still know what this is about.

Please have a think. This is not a very calculation-heavy question. This is something you should know or could engineer from your formula sheet.

So let's see what you think. Who thinks it's answer A? Answer B, answer C, D, E, F, G, H, and I.

Okay, you voted for three answers. Answer F is correct.

The speed and the medium changes, so the wavelength has to change. The wavelength becomes smaller. So we have V , speed equals the wavelength times the frequency. If V is lower, frequency is not changed and λ has to be lower as well. So inside a medium with a higher, or with a refractive index higher than 1, the wavelength is shorter than the wavelength and value.

And that's actually what we did. We looked at the nature of light last time, reflection and refraction. In two ways, basically, we used Snell's laws, which you all probably love and hate because of the trigonometry involved if you do calculations, but also Helgen's principle, which could also be used to explain the phenomena we saw. And we looked at total internal reflection. Well, let me show you a nice movie of some phenomenon that we now should be able to explain. I think it's this one.

Yeah, that's the one. Let's see if you can judge what happens. Where's the glass going?

It's weird, right? It's still there.

Does anyone have a clue what is happening here?

You're kind of on the right track, so you say there's no change in angle. Why would there be no change in angle?

If the refractive index is different, then there should be some bending. Ah, you should have started with that. Indeed, the refractive index of oil, it's vegetable oil and pyrex glass. So the refractive index is the same. So it's optically impossible to distinguish between the different media. They have the same refractive index. So for light that we see, it just goes through unimpeded without bending at all. So indeed, it's caused by the refractive index being exactly the same.

So you can't distinguish the materials. Okay, today on the menu we have dispersion, polarization, scattering. So that's kind of wrapping up chapter 33. I'll talk a bit about rainbows as well. And then we look at bits and pieces of chapters 35, 36, interference and diffraction. Those are also concepts we will need for quantum mechanics because we will do the two slit experiment in detail there. So first dispersion.

Dispersion is a, well, it's a general name. It's anything that is frequency dependent is called dispersion. So what you see here is the refractive index of

various types of glass and how it changes with the frequency of the light as we see it. So a typical glass, well, let's do quartz.

It has a slightly lower refractive index for red light and a bit of a higher refractive index for violet light. And this means that in a prism we get spreading of the light. So it depends on the color of the frequency. This does not end in, well, at the visible spectrum. This goes throughout the electromagnetic spectrum. It's just at the visible light. We can see it most clearly. And the result is that different colors can and will reflect at different angles. So that's why when you get these pictures. And conceptually for red light, it's always, it's always the case, always the case that the refractive index is lower. So it will bend less towards the normal than violet light if you go from air to glass. So that's also what's happening here.

Red light is bent less than violet light. So we get this basically decomposition of light into its colors that are separable. This is a continuous spectrum, of course. What comes out is a continuous effect. It's just that we have defined certain wavelengths as being a certain color. So this is continuous. It's not discrete in these bands, but we have declared these bands to be of a certain color. And that's why we, well, we kind of, our eyes are trained like seeing different colors as separate colors. But it's a continuous spectrum, of course. And this also happens in rainbows. So there's lights coming in from the sun. So you always have to stand with your back to the sun, with your eyes facing where the rain clouds are. And then sun rays are coming in and they are, well, reflected twice for the primary rainbow. I'll come back to that in a second. And then they meet your eye. So the sun should preferably be quite low and the rain should be in front of you. And then you get effects like this.

So array of non-separated light is coming in. And then the red light is bent less than violet and blue light. This happens twice. So there's twice reflection within a raindrop. This is a spherical drop in vacuum. So an idealized situation. In practice, it's slightly more complicated. It's not exactly spherical. But you separate the colors being, well, red light reflected less than violet light. But this feels weird, right? If you look at a rainbow, what color is on top?

Red. Indeed. So this picture does not make sense completely yet. I'll get back to that in the next slide, actually. So light comes in. It's reflected twice. And the angle for red is of this angle, because it's reflected over less of an angle. This angle is larger than the angle for violet light. And there's also total internal reflection involved. And due to the dispersion, these colors are split.

And arrays of color, different color, generate from different positions of the rainbow. And that's why we see red light at the top. So what happens here is, at the top, we have the light that is reflected over the largest angle. So red light is reflected over a larger angle. So to see this, well, these drops have to be higher up. If they are higher up, then this angle makes sense. So higher up, there's a larger angle. That's the red light we see. And the drops lower down, they are reflected over a smaller angle.

And that's why we see blue light and violet light from there. So we don't see the rainbow as being from one drop, given all the colors. It's depending on what height these drops are, that we see a different color. All right, so that's often misunderstood. This has been an old exam question in multiple choice sections.

These drops at the top give you the red light. These drops at the bottom give you the violet or blue light. And the spectrum in between comes from the drops in between.

This is a subtle detail. Physics still works out, but it comes from the fact that

the light is, the red light is reflected less than or bent over less of an angle. So this angle is larger than violet light, which is over a smaller angle. So it gets bent more. So it has to originate from a different position in the sky.

Okay, this is the primary rainbow. There's more.

So this light comes in and is now reflected. Well, there's twice internal reflection and then refraction again.

And now the colors are reversed. Because now white light is coming in, it's separated into its individual colors. And then red light is now at the top. Again, this situation holds that these lights, well, the colors you see come from drops at different height. But now the colors are reversed. If you compare this picture with this picture, here there's refraction, total internal reflection. Refraction again, red at the bottom. Here it's the other way around. It's refraction, total internal reflection twice. And then refraction again, such that we have red light at the top here. It makes a longer path through the drop. So also the rainbow is a bit fainter than the primary rainbow. Bless you. So that's typically what you see is you have the primary rainbow with red going to violet and then the secondary rainbow inversely colored, red and violet on the outside. And it always seems that in between it's darker and that's caused by the fact that it feels like the lighter side is always on the violet side. And the violet side is on the outside here, violet side is on the outside here. So in between it seems darker.

You can have higher order reflections as well, but these are becoming so faint that you can't observe them anymore.

This can happen with rain, but it can also happen with fog, a fog bow.

But the effect is very, very weak because fog droplets are much smaller than rain. So it is what's more difficult for light to go through this process of twice refraction, one times internal reflection.

So there's very faint coloring on the fog bow, but it's still possible.

More and more examples are sometimes you see if you look at lights on the sun, you see a halo around it and that can be caused by ice crystals in the sky.

And ice crystals typically have a non-circle shape.

They are kind of hexagonal in shape, which means that light can just refract twice over such an ice crystal.

There's no internal reflection needed. And in that case, well, red bends a bit less.

So these droplets come from, sorry, these colors come from ice crystals which are more towards the center. There is the red light. And if you go more to the outside, you will have the violet light.

Due to the shape of this crystal, there's specific angles involved.

So this happens at a 20-degree angle around the sun or a light source because of the geometry of the ice crystals. But the effect is still, well, similar. Due to dispersion, the colors separate because the red and violet and all the colors in between, they refract over different angles when they go through the ice crystals. Often, when the sun is quite low, well, this happens if the sun is over top. You look straight up. If the sun is quite low, it's still the same effect, but it's, well, amplified in horizontal direction.

And then it looks like you have multiple suns, sun dogs or parahilia, these are also called. And this is mainly caused by the fact that the ice crystals are oriented vertically, such that the refraction is in horizontal direction.

So the effect is still there.

You can see the, well, the halo still. But, well, with this low sun and with the ice crystals mainly oriented vertically, you have the effect amplified in horizontal direction.

These are mainly examples. What's important is that you understand this concept, that light from different colors comes from different heights of water droplets.

Are there questions on this so far? We've all seen rainbows, right? It was a lot of rain last night.

No sun yet. So try to summarize this for yourselves and then we'll continue with polarization.

Just to clarify a bit further, due to one question I got now.

This effect happens everywhere. There's ice crystals all over the place, but only at specific points in space. It satisfies the condition that light comes in and bends towards us for red light at an angle of 21.54° . So exactly at these points in space relative to the light source, we meet this condition that red light can reach our eye slightly further outwards here.

We have that same condition for violet light or blue light. So ice crystals are throughout. This happens everywhere, but just only at these spots we see the effect. We see the color separate and the same holds for the regular rainbow. Well, in this case, ice crystals are everywhere again. So also relative here, we meet this condition.

That is also caused by the fact that the sun has a point source. So we see the point source is behind you. So, well, let me demonstrate how it works. The sun comes like this and at this point here we see drops reflect. We meet exactly the condition that they reflect red light to our eye. Also here, because the sun is really well here at this point. You understand what I mean?

It's hard to explain the words, but I hope I made it visual to you. So there's a point source behind you, the sun. Because you look at the rainbow from a single point of view, all of the points that meet the condition for the red light to reach your eye are in a circle in front of you, in a semi-circle. We would see a full circle. Yes, yeah, yeah. Luckily we have ground, otherwise it gets messy. More questions on rainbows on this person?

Sorry? At the end of the rainbow, yeah, you'll find out after the exam.

Hopefully you pass. Okay, another quick recap question.

So this is on total internal reflection. Let's see if you still remember what that was. Which is also the concept we used for rainbow, right?

Under what conditions does this happen?

And sometimes it helps to make a drawing.

So we're talking about this situation, θ_A , θ_B .

When did this happen? Well, this is something you can engineer from Snell's Law, right? Let's see if we can vote already. Who thinks it's answer A, answer B, answer C, answer D, and answer E?

Most voted for answer A, but also some Bs and Cs. It is answer A, actually.

So we need refraction away from the normal. So we go from a higher refractive index to a lower refractive index. So from medium A to medium B, n_B needs to be lower or n_A needs to be higher than n_B . So that is this one or that one. The rest is excluded already. And then we need an incidence angle that is at least higher or more than this angle than this happens.

Then we go from medium A to medium B. So this is a very nice mechanical example I'll show in a video. But polarization happens for transverse waves only.

You can do it for mechanical waves as well. So let's start with mechanical waves and then build our way to electromagnetic waves. So if you have a wave propagating on a string, then we have a vertical excitation. A wave is propagating forward. So it's a transverse wave and it's polarized in y direction. You can also have this in x direction or you can whirl it around in a circular sense and then you get a circular polarization with both x and y components.

Sorry, both, sorry, my bad. We always have z as propagation direction. This is x, z polarized, this is y polarized. Propagation direction is x. We could select a single polarization by putting a barrier after it. So with a slot only permitting y amplitudes, we could select the y amplitude of a wave that is arbitrarily polarized. And then we call that linearly polarized.

So linear polarization, there's only one direction of displacement. So even if you whirl this around in a circular sense, you select the y polarization. And this can be done by a so-called polarization filter or polarizer. It only lets waves with a certain polarization through. To show you what that looks like, let's do this one.

Let me show you, let's see, mechanical waves. I can't get the wave through.

No matter what crazy wave I can, only the left and right components get through. It's been polarized. Now that only works on transverse waves. If it works on light, then light is a transverse wave. So this is an effect you've probably seen also in sunglasses. It can get polarized.

So light gets polarized by a polarizing filter. So it has to be a transverse wave.

And as you probably know, light consists of transverse electromagnetic waves, which have both E and H coupled with E and H are perpendicular to propagation direction. At least here is correct. We always use that for propagation direction. And you can have a forward or backward traveling wave, but both E and H are perpendicular to the propagation direction, which is Z. And polarization is in this case defined as the direction of the electric field. So let's also look at that.

So there's a dipole antenna oriented like this.

It generates electric fields vertically. And also the receiver is a dipole antenna. Try to formulate an answer. I'm going to see what happens. So that's something you will learn in the follow-up course, electromagnetics. For now, we stick to light. There's a question. Go ahead. No, it doesn't.

There are some ranges over which it works. And that's also why if you look at the microwave, people have a microwave at home or in your student home. There's in the

glass, there's a very fine grating and that prevents all the radiation, the electromagnetic waves of coming out of microwave.

The higher frequency, the smaller you need the mesh to block it, but it works over a larger range of wavelengths. It doesn't have to match the exact wavelength of light. It works over a larger range.

So light is an electromagnetic wave and that's the reason why we can polarize it. And if you have light from an arbitrary source, the sun or a light bulb here, then it's not polarized. It has all kinds of polarizations in it. And if you want to do experiments often, you need light polarized in one direction. And to do this, you can put a polarizer behind it or a Polaroid filter, if you wish. So there's an unpolarized light coming in.

You have a filter which is polarized in vertical direction and it lets through only light in vertical direction. Just as the example that was shown in the movie, the first movie I showed with the black sheet of polarizer. And after this, you have polarized light.

So one polarized component is absorbed and the other polarized component is passed on. If you put two of them, one like this, one like this, nothing gets through. If you put one like this, one at an angle and a one like this, then due to quantum effects, something can get through. But I'll get to that later in the next module actually. But if I have one like this, one like this, no light comes through. And this is something you've seen in sunglasses, for instance.

I'll get back to that in a second. So after such a polarizer, the transmitted light only has one polarization. If I put this at an angle, I can choose the angle at which we have our light falling onto a, well, a measurement device, a photocell. A natural light has, well, contains all polarizations and it is kind of uniformly spread over those polarizations. How much light is there per polarization?

So it's equally distributed, uniformly distributed. And after polarizing, we select only one direction.

Since it's equally polarized here and it's uniformly distributed, after polarizer, only half of the light is left. So it gets darker after polarizer because we've, well, we've gotten rid of half, half the light because that is absorbed in the polarizer, the rest gets through.

And that's something you've seen in polarized sunglasses. So this is the same material as was shown in the, well, in the sheets that was held up against the light. It's darker after a polarizer because half the light is absorbed. And if you stack them with the same polarization direction, again, half is absorbed. So one quarter is left here.

Question? I'll get back to that in a second. There is some special cases when polarization can, well, change due to reflection. I'll get back to that in a second. So if you have them in the same orientation, you just reduce the signal basically by factor half, factor half. So we have one fourth left. And if you rotate it by 90 degrees, well, you've selected the vertical polarization.

And if you then put the other glass in the other direction, you've basically killed all the light that is going through. We have complete blackness unless you put a 45 degree angle polarizer in between.

This is probably something you've seen before, right? I don't know how sunglasses work. Polarization helps also. And I'll get back to that in a second.

First, there's a physics, well, concept that describes how much light is coming through. And that's called Malus's law.

So if you have a first polarizer, you select only the electric field in this direction. So half of that is coming through, half of the incident light. And if you then put another polarizer at an angle, ϕ , relative to the first polarizer, you can tune how much light is coming through. If you put it in the same direction, ϕ is zero, then everything gets through. If you put it at a 90 degree angle, ϕ is 90 degree or π over 2, then nothing gets through. And this, well, this feels like a cosine law. If you angle zero, you have a factor one that is transmitted, angle 90 degrees, then the factor zero is transmitted. So it should behave like a cosine. When the intensity of light is proportional to the electric field squared.

So the intensity we get after the second polarizer relates to cosine squared of ϕ .

So the light intensity we get after the second polarizer is cosine squared of ϕ times the incident light.

And this is called Malus's law. So you can stack them after each other. The effect is that you get a cosine ϕ factor after every polarizer. And the intensity is under square of what you have left. Typically, we work with two, because then in that case, you select a linearly polarized direction, and then you can tune how much of that is coming through.

Is that clear? And of course, after the second polarizer, you are polarized in the direction that you have left. So the second polarizer is vertical in this case. So what remains is only a parallel component here, vertical component, which is $\cos \phi$.

That's the question. That absorbs the directions that are not transmitted.

So it really absorbs the polarization that is not transmitted. Yeah, so it is blocking through absorption, basically.

Questions on this? Yeah, go ahead. No, it blocks the electric.

Well, yes, indirectly. Let me rephrase. Yes, it is affected because in an electromagnetic wave, they are coupled. If you block one, then the other also decreases. Because E and H are always coupled. If you propagate in that direction, E and H are like this, or like this. If you kill one of them, then the other one also dies. They are coupled through Maxwell's equations. So you can't tune them independently, at least not if you have a wave. If you look at statics, then you can have a static magnetic field without electric field, or vice versa. But if you have waves, then they are always coupled. If you block one of them, you could also block H , and then the electric E would be blocked as well. But they are always coupled. So it's sufficient to block only one of the two to kill the signal. And that's how a polarizer works.

Intensity or polarization is defined with respect to the electric field. So we look at the electric field, but it implies also the magnetic field that it's reduced by the same factor.

There are special cases when there is reflection of light.

And then due to reflection, it gets polarized. And that's probably the case you refer to or thought of, at least. If there's light coming in onto a surface with a certain refractive index, then light can be, well, it's a transverse wave. So the wave is propagating in this direction. So it can be polarized either in this

direction or in this direction. So both E feel like this or like this, but it's propagating in that direction.

At a specific angle, the reflected ray does not contain this polarization anymore, just the one that is out of plane. This happens at a very specific angle where this is exactly 90 degrees.

So when the angle of incidence and the angle of refraction form a 90 degree angle, and then you polarize by reflection. And this is called the Brewster angle.

So at this specific angle, I'll have a detailed derivation on the next slide. At this specific angle, there is, well, one of the two polarization directions disappears.

So the vertical polarization is gone. Only the horizontal polarization, which is, if you consider a vertical interface between the media. So this is the air and this is stable. And A is air and B is a table. Then light polarizes like this, reflects like this. And then after reflection, only horizontal polarization is left. So how does that work? This is called the Brewster angle. As I said, this happens at a very specific angle.

If θ_P plus θ_B is 90 degrees. So these three angles together have to be 180 degrees. This is 90. So θ_P plus θ_B has to be 90 degrees. So this happens when θ_B is 90 degrees minus θ_P . Well, we can do Snell's law for this.

So we know that θ_A has to be θ_P . So let me do that actually.

We have in general Snell's law, $n_A \sin \theta_A$ equals $n_B \sin \theta_B$.

We know that $n_A \sin \theta_P$, that's the incidence angle, θ_P .

And then θ_B has to be 90 degrees minus θ_P .

So that's $n_B \sin 90 \text{ degrees minus } \theta_P$.

And this I can write as $\cos \theta_P$. So I get n_A over, sorry, n_B over n_A equals $\sin$ over $\cos$.

So that's $\tan \theta_P$. So I divide by n_A and I divide by $\cos \theta_P$ to get this expression, which is then Brewster's angle. So this happens at a specific ratio between, well, the refractive indices. This happens in both cases.

The tangent can take from zero to infinity. So it doesn't matter if you go from high to low refractive index or vice versa. This can happen for any ratio and only at this exact ratio where θ_P equals this.

$\tan \theta_P$ equals that ratio.

So contrary to the critical angle, this happens for any angle larger than or equal to this critical angle. This happens only at this exact angle.

So that's the difference. So otherwise it would be, well, physics just sort of works like that.

So other questions on this? I'm not sure if this was covered in secondary school questions here.

So you both get a refracted and a reflected ray, but the reflected ray only loses

one of the polarizations.

So what remains is just a horizontal polarization. So you have polarization like this, a polarization like this, and only this one remains after. This one is gone. So only at this specific angle of incidence, the reflected ray has one polarization less. And I think that's a nice moment to do the break. Please think of this concept during the break and we'll have two quick quizzes after the break and also the Mentimeter quiz.

Please take your seats again.

I think I'll do the Mentimeter quiz at the end of the lecture. I think that makes sense. It will not disturb the flow too much. It would disturb the flow too much now. That's why I'll do it later. So let's see if you understand how sunglasses work.

Not the cheap ones, which is just colored glass, but the ones with the polarized glass, polarizing glass. So which polarization would sunglasses filter out? Perhaps you know, perhaps you don't.

But there is a reason for their direction.

Well, let's vote already.

Answer A. Answer B. One or two. Answer C.

Answer D. Okay. Why answer D? That's the correct one.

No, it's not that reason. No, actually it has to do with Brucer's angle. Due to Brucer's angle, there's a lot of reflection of sun already on any surface which kills the vertical polarization. So most of that is filtered out already.

So actually what happens, let me go to this here. So a lot of light when you're driving a car, for instance, there's light coming in. Coming towards you.

And then a lot of this has satisfied this already. So what's left is mainly the horizontal polarization. So the vertical polarization has been reduced already by Brucer reflection.

And that means that the sunlight you see reflected on any surface mainly has horizontally polarized light or has a dominant component of horizontally polarized light. So that's the part that needs to be filtered out. So the sunglasses are transparent for the vertical polarization. Horizontal polarization needs to be filtered out more because that's what's still left in the light.

Another question on Brucer's angle itself. Let's see if you understand this.

So let's vote. Answer A. We have a question. Okay. Let's first answer the question on this. Then let me do this one first. I'll get back to the previous question. Answer A.

Answer B. C. D. E. Okay. You paid attention well. The refracted light, that's the one that's going into the medium, will still contain both polarizations. So that happens here. The refracted light still has both polarizations. It's a reflected light that only at exactly the Brucer angle has one polarization left. But the refracted light still has both.

So that only affects the reflected light, not the refracted light.

So back to your question then. If you have two polarizers at 90 degrees after those two, you have no light at all.

Then it's going to be horizontally polarized. What is the question exactly?

You wear them like this always, and they're polarized such that vertically polarized light gets through.

No, but this is how they are designed because most of the light that reaches your eye has a dominant component of horizontally polarized light. So that's the one you need to filter out because that could lead to blinding or to issues with the driving. What's the question there?

Brucer's effect is only happening at this angle.

So if you're outside, there's a lot of things that satisfy exactly Brucer's angle. It's just a bit less vertically polarized light in there. Because you see light coming from, well, if you're here and the sun is here, then there's light coming from reflections here, here, here, here, from any reflection. Only a small range here will satisfy exactly Brucer's angle.

The rest does not, but there's slightly less vertically polarized light in there because of Brucer's angle. Yeah, you will see a bit higher intensity probably because there's also other polarizations being filtered out. In view of time, it will have to continue, otherwise we will not make it through. Yeah, this one I will show.

Also a nice effect. There is a single red dot and a white piece of paper.

The red dot appears as two separate dots. This is because the two separate polarization components perpendicular to each other are refracted in two different directions by the calcite crystal. This can be shown with this piece of Polaroid.

There is one in the rays, one of the images of the red dot, here's the other.

This is called birefringence for double refraction.

So what happens here is that due to the crystal structure of calcite, so this, this salt-like material, there's two directions in the crystal that polarize light differently. And that's why when you place it on a dot, you see two dots and then the fact that it polarizes differently is shown by this Polaroid filter and that's on top and then you see only one of the two dots away.

Yeah, yeah, I will still manage.

So scattering, that's the next topic. All of this still relates to light, still relates to 33.

If there was no scattering due to air molecules, we would see the sky as black. We could always see stars, but due to scattering we see the sky as blue. Actually it should be violet, but our eye makes it blue and I'll explain that in the next slide. So we know that sunsets are red, sky we see as blue and this is caused by scattering due to air molecules. Scattering is an effect that relates to molecules that absorb light and then re-radiate light. This re-radiation happens in all directions, so light that is absorbed and re-radiated will have less intensity on this side. And due to, well, the specific size of the molecules, this happens at certain frequencies, you can see certain wavelengths.

Scattering is the effect that is absorbed and re-emitted, re-radiated and this

relates to the size of the particles in comparison to the wavelength of the light.

So it, well, very small particles cannot absorb long wavelength light, that makes sense. You have to be in the same order of magnitude as the wavelength to be able to interact with the wave.

A lot really already demonstrated that due to scattering there is an intensity decrease that scales with one over λ to the fourth. So if you think about two types of light, the two ends of the visible spectrum, you have violet light, 380 nanometers, red light, 750 nanometers. If you put that in and calculate the ratio between the two, then there's a difference in intensity that's a factor of 15.

So this means that, gee, let me say it correctly, violet light is scattered more.

So if you have a smaller wavelength, then this fraction is higher. So one over λ is larger if the wavelength is small and that to the power four. So violet light decreases more due to scattering than red light.

So the light you see, well, depending on where the sun is, you see more red or more violet light. And I'll explain what that means in the next slide. So scattering of blue and violet light is 15 times higher than for red light. That comes from this ratio. And therefore sunsets are red. If the sun is low on the horizon, there's a long path through the atmosphere.

And then all of the violet and blue light has been scattered already, that is gone. So the intensity of that light is reduced more than that of red light. What remains is the red light, because that's scattered less than violet light. So that's why a sunset is red, only the, well, oranges, orangish or red, only the dead light remains. The rest has been scattered already due to the longer path through the atmosphere.

If you look straight up, if the sun is high up, then you would expect the atmosphere to be violet because that scatters more than the path through the atmosphere is shorter because the sun is straight up. And then you would expect this guy to be violet, but it is blue actually. And that is because of, we don't see violet very well. Our eye filters, well, as these three cones on your retina that select light, so it's a yellow or reddish one, a green one, and a blue one. Violet, we don't see really well. So what remains, our eye, or kind of makes that blue, because that's how our eye works.

That's why the sky is blue and not black because of scattering.

There's air molecules that scatter light and the dominant effect, if the sun is high up, is the blue one. If the sun is low on the horizon, then we have red light mainly.

But let me get to this one as well. So the sky is blue, but clouds are white. Clouds have water droplets.

They are larger and more dense than air molecules. And that means that they scatter more light and they have a lot of scattering that happens. This happens kind of uniformly for all colors. And that's why we see clouds as white. So this is kind of independent of the wavelength because of the larger and more dense water molecules compared to air molecules. That's why clouds mainly show white.

So this is independent of wavelength.

Not a lot of dispersion. But if the sun is low on the horizon, the light makes a long path through the atmosphere. All of the violet light has been scattered and

what remains is just the orange and the red light of the sunset.

Try to summarize this for yourselves. And then we go to all the next chapters in the book, basically, with some nice examples. There's still a lot of ground to cover. Let's do a one-minute recap and then we continue. So let's look at the next topic.

I'm now going to talk about interference and diffraction and also the combination of both eventually.

Interference is a concept that must have been treated in secondary school, probably for sound sources, for instance. I'm not sure if it was covered for light. This happens for any case in which there's more than one wave source. So I mentioned superposition when I talked about waves, wave propagation. Interference can happen in any situation where there's two or more waves overlapping. So there's two droplets in a pond or in a bowl of water. And then where the wave fronts combine or superimpose, you can have interference. You can have constructive interference or destructive interference. And interference is typically thought of if there's a small number of sources, typically two or three. If it's more than we more often talk about diffraction for reasons that will become clearer in a second.

So the superposition helps us with understanding this. If you have two or more waves overlapping than the combined effect, the resultant displacement is then the superposition of the two instantaneous displacements from any of the sources. So you can sum the individual effects. Superposition is really a useful principle. That's something that holds throughout physics, it holds for Maxwell's equations. As long as the equations are linear, you can do superposition. In fluid dynamics, if you have Navier-Stokes equations, then you can't do that anymore. But for any regular physics that we cover in this course, superposition holds. So how do we go about with this?

Suppose you have a source, which could be any wave source, can think of sounds or waves propagating outwards. On a water surface, we typically assume that the wave source has a single frequency, and therefore a single wavelength. So we can make these wave fronts for a point source in 2D. Those would be circles. And such a picture is then a snapshot of what these wave fronts look like for this situation. So there's a wavelength and they propagate outward at a certain speed, v , that's f times λ .

If you have two sources, two identical sources of monochromatic waves, waves of a single frequency, and you place them apart, then you get a superposition pattern, an interference pattern, that locally, for instance at point B, has waves in phase. So there's two wave fronts, the green one and the blue one crossing here. So locally we are at the same phase here. So we get constructive interference for point C.

So in between the green ones, but at a crest of the blue one, we get destructive interference.

So depending on where we are, relative to where the sources are, we can get different types of interference, constructive interference or destructive interference.

If we assume that the sources are in phase, so they radiate out waves at the same frequency, then we get these wave fronts and then we can construct it like this.

There's a question here. In what sense not identical?

If they did not have the same frequency, then there would be a very complicated

pattern.

Then still superposition would hold, constructive and destructive interference would hold, but this pattern would also change over time. If the two sources are in phase and are the same sources, so at the same frequency, then you get kind of a stationary picture like this. If they are not at the same frequency, this picture shifts over time. It would still hold, but it's difficult to calculate that. So in questions you'll solve on this, I'll state that we have the same sources or monochromatic sources, sources of a single frequency, which may or may not be in phase. But if there's a fixed phase difference between them, then you would still get a stationary picture. It's just shifted. Either the green wave fronts or the blue wave fronts would be shifted in the propagation direction over a part of a wavelength, but you would still get a stationary picture. So at point A we are at two crests, both for the blue one and the green source. Now we have constructive interference, and if you are at point C for instance, you're in between the green ones on a blue wave front, then you get destructive interference. So you're exactly out of phase, so a local minimum you would get.

So when does this happen? If the distance between points is a multiple of a wavelength, then you would get constructive interference. If the waves are in phase and then they arrive at point B, then you have a distance of seven wavelengths here, nine wavelengths here. So the difference here is two wavelengths, exactly two wavelengths, then you are in phase.

The wave from source two will be a bit weaker because you have propagated further, so you will not cancel out exactly, but you are at a local minimum in this point in space, because there is an integer multiple of wavelength two, in this case between the two sources.

You would get destructive interference if you are m plus half wavelength separated, so if there is half a wavelength difference between the two. So at point C here, the distance to source one is nine three quarters wavelength, and source two is seven and a quarter wavelength. So the difference is two and a half wavelengths, so whenever there's a maximum coming from source one, there would be a minimum coming from source two at this point, and then you would get destructive interference. Again, not completely, because the difference, the distance to point one is larger than point two, so the blue wave has, well, decayed a bit more than the green wave, so it's approximate cancellation here, but this stays like that, the waves are out of phase. And that's also why, if you go back to, and get back to your question in a second, if you look here, exactly in between, there's real cancellations, but if you go further away, then the cancellations are not complete anymore, because you have propagated further away from one source than the other, so you get approximate cancellation. But exactly on the line in between, you get this, or exactly, in the area exactly in between, where they have traveled the same distance, you get really exact cancellation, or amplification.

There was a question here. No, we will not look at decays of waves.

We assume that they stay the same, otherwise it gets too complicated too easily. We will look at, questions on this, for instance, have been multiple choice questions, saying that, okay, at this point, will you have constructive interference, destructive interference, or no interference at all? Or what conditions do you need to meet to have constructive interference here, and destructive interference there? So, so far, I've only done conceptual questions on these in the final test.

So this is the same condition, but shifted over half a wavelength. Here it's m times λ , so an integer multiple of wavelengths.

Here it's m plus half, so a half integer, so m plus half multiples of wavelengths.

And the complete picture then would be like this, if you have sources, like we had in the picture, the blue one, the green one. Then, depending on where you are, you have local maxima and minima of, well, m equals zero, so a point here has no wavelength difference. They are exactly in phase, same distance, so m is zero. At this point, there's one, two, three, four, five crests of green, and there's one, two, three, four crests of blue, so there's one wavelength difference, etc.

You get constructive interference, and in between, you would have the nodes where there's destructive interference. Those lines would be in between the red ones.

Where is the MS4 line? Straight down, yeah, so they are separated, four wavelengths apart, so the MS4 line would be straight down. So that curve is basically not shown, so it's straight down only. So there's no more antinodal lines than these in this situation. And the further away you get, well, the less the effect becomes because of a difference in distance. So for the MS0 line, you have exact cancellation or amplification for MS1, two, three, it gets less the effect, because, well, the wavelength difference translates itself to a decay in amplitude.

Is this something that was covered in secondary school or not?

I see people nodding, that's good. This also works for light, and this was done by Thomas Young in 1800, where there's two sources, monochromatic light, it's led through one slit, and then this slit acts as a source, and then it falls on two next slits, and each of these slits acts as a source for light propagating in a semi-circular way, and as a result you would get an interference pattern on a screen that is far away.

So this was a two-slit experiment by Thomas Young, and if you would do the calculation on that, well, the actual geometry would be this, then you have the two slits that are distance d apart, and then there's a screen that is at a distance r , a finite distance r , and then you would have a slightly more complicated pattern than if you do what is typically done in literature, you're approximated by a screen that's far enough away compared to the distance here, that you can have these beams as parallel. So this is how you derive the formulas, in fact it's slightly more complicated if you have a finite difference where the screen is.

We will use this one, and what you can see here is that if this $d \sin \theta$, that's the angle at which the beam array comes out, if this is an integer multiple of wavelengths, then there's an integer multiple of wavelengths difference between these two arrays, then you would have constructive interference, and if this $d \sin \theta$ is m plus a half times a wavelength, you would have destructive interference. So that's basically what happens if this path distance Δr , which is exactly this edge, if that is m times λ , you would get right regions on the screen, so constructive interference, so $d \sin \theta$ equals m times λ , and if it's m plus a half times λ , you would have destructive interference, so dark spots on your screen. So that's how the double slit experiment works. This is probably also covered in the secondary school. I think the double slit was already in the curriculum back in my time, which is over 120 years ago.

Double slit experiments. So I think in view of time, let me continue. This is something you've seen, right?

Let's go to diffraction after a quick quiz question.

Let's see if you understand the concept. What happens if you change the wavelength of light? Let's see, are we ready to vote?

Who thinks answer A? Answer B? A and B both chosen.

Answer C? Color changes. Answer D? No one? Answer E? None of the above.

The correct answer is answer B, and that follows from other formula for interference.

If the wavelength of the light increases, λ increases, so the spacing then in the pattern, the sign of the angle will also increase. D is the distance between the slit, that doesn't change.

m is the counter for the constructive and destructive interference. If λ increases, then $\sin \theta$ has to increase, so the angle has to increase. So the light parts of the pattern move further apart.

So that's why the bright areas move further apart.

Okay, last topic for today, diffraction. It's closely related to interference, but it happens if there's many sources.

So if you think about geometric optics, then if there's an object and you shine light along it, then you would get a sharp edge on a screen that's somewhere. So you would get a sharp shade basically of your object, shadow.

But due to the wavelength of light, this edge causes diffraction.

This causes wave propagation patterns that blur the edge of the shadow.

This is called diffraction. So what happens if you make a photograph illuminated by monochromatic light, so the effect is reduced when you have light of multiple frequencies, but if you have monochromatic light source, you would see blurry edges here. And these fringe patterns, so these blurry edges, also these fringes here, are caused by diffraction. And you can use Huygens wavelets to explain this.

So every edge on the object will act as a Huygens source, and with that you can construct what happens. And this happens if you have a lot of sources. For the two slit experiment, there's clearly two sources. This happens in general for any object, but it also relates to the wavelength in comparison to the object.

Can all waves undergo diffraction?

Can sound waves undergo diffraction, for instance?

We think yes, we think no.

The answer is yes. Can you give an example? Yeah, if there's an open door, then that would be caused by diffraction. It could also be sound propagating through the wall due to the sound exciting the wall. But if there's an open door and you're not standing in front of the door, you can hear someone on the other side of the door talking. That's caused by sound waves going through the door and then spreading out again. And that's diffraction.

Or if you're standing around the corner of a building and you hear someone around the building. So that's also diffraction.

So examples are here, where the waves are coming in, I think in this direction. And due to the sharp edge, this point acts as a Huygens source. So the wave propagates in all directions. Whereas traditionally, if the wave is coming in from this side, you would have a dead zone basically for anything above this region. Due to diffraction, waves also propagate in this direction.

Two more examples. Probably you don't even know what this is anymore.

I'm not sure about this generation, do you? I'll keep it in for many years and see when people really don't recognize it anymore. This is caused by diffraction by basically the pixels, the colored pixels on your phone. Where also they get, they give an interference or a diffraction pattern basically. Of course, by light reflecting on the different pixels. And here on a CD, there's all kinds of bits on the bottom side that store the information. And the combination of this pattern of bits can also give diffraction. And this, well, there's not monochromatic light falling on this. So you also get dispersion to see this rainbow pattern. So how does that work? So instead of a single slit, which would give a single slit image on the screen, you get a pattern that looks like this. So if the condition for the slit with respect to the wave length are satisfied, then you see a diffraction pattern with a center, bright region, and then there's dark spots and then again, light regions. And the construction is similar as what we did for interference pattern. But now we have, well, we call it Huygensworsers now, because we have a lot of sources. If you would zoom in on a slit, you could make any of the points here in the slit a Huygensworsers. And the effect of that, that's called Fresnel diffraction in the book, the effect of that is that if you have all of these sources, you could get constructive interference at a point P on the screen far away. Question?

It would spread out a bit horizontally. Typically what you would see is you would, this is an idealized picture, what you would see is also here and here and here, you would get repetitions, but the dominant effect is often caused by the smallest direction of the slit. That's where you see the biggest effect of the pattern. It would also repeat horizontally, but only in these two perpendicular directions, because that's how the slit is oriented. For a circular, or not slit, but hole, you would see a spherical interference pattern, a diffraction pattern, sorry. There's examples of the book, in the book in which you, there's a circular hole, you would see a center bright spot with circles of shadow and light around it.

Yeah, that's also caused by diffraction. Yeah, we are talking about monochromatic light, so light of a single frequency. If you have just a triangular hole in a sheet of paper, we have light of all colors and all of the effects get mingled. I haven't tried that before, but interesting thought experiment.

No clue yet. I guess there will be diffraction, because that's how physics works, but what the exact pattern would be, I'm not sure. I'll try to find out for next time.

Yeah, in view of time, let me continue this. Yeah, I'll manage.

So this can happen in, it's called Fresnel diffraction. If it's near field, typically we again look at far field considerations. You can use this for imaging as well.

Microscopy can also use this. We typically look at this case, where there's a screen far away, and we again look at constructive or destructive interference. And this again relates to if there's integer multiples of wavelength difference between the path that these light rays have traveled. So what does that look like?

Similar picture as we had earlier, but now we have a continuum of sources here, not just two sources, two slits, but a continuum of sources. And again, we look at the distance here.

If that is an integer multiple of wavelengths, we get constructive interference. But since we have a continuum of sources, basically, so this is half the slit. For half the slit, we can already set this calculation up. The first dark region would

be if there's half a wavelength difference between the paths that these rays have traveled.

So when that happens then, well, integer multiples of half a wavelength.

So this comes down to two λ over a . So what this is the derivation of the formula, what you typically use is the following. So the Fraunhofer diffraction pattern, the far field diffraction pattern, has one central bright region, and then you have local minima, where there's destructive interference.

So we have a series of dark fringes separated by light fringes. And this happens at sine θ is m times λ over a . So at m is one, you have the first dark fringe, m is two, the second dark fringe, etc. So the definition for diffraction is different from interference.

Because here we started m equals one. For interference, we started at m equals zero. But it's defined different here because we have the location of the dark fringe for interference. So that's something you should keep in mind when working with this. And often for light, we have that the wavelength is much smaller than the slit size. So the angle, and that's the approximation you often see made in literature, or in physics books, is that sine θ is approximately θ because the wavelength is much smaller than λ .

So this is a small number anyway, so it doesn't matter if you take the sine of the angle or not. So what does that look like? Well, to summarize, the intensity follows in this profile.

There's diffraction here with one center bright spot, so this band, and then the intensity follows a certain profile that you could set up a formula for, but we don't do that. And it's not needed for the conceptual understanding, which is enough for this course.

Are there questions so far? I'm running out of time, so let me, I think we'll start the break a bit later and then we can still do the man-to-meter quiz. Well, still people are here. There's people leaving before to your classroom typically, but let's do the man-to-meter quiz in two minutes.

So to summarize, the diffraction intensity depends also on the width of the central bright beam.

Depends on the, well, we look at M_{S1} . So the width, the first minimum is then a λ over A , and if A is λ , you get a very wide center beam, but if you have a slit that's larger than the wavelength, or if a wavelength is much smaller than a certain slit, you get a really a focus of the intensity towards the center, well, the center region.

The wider the slit, the more narrow this diffraction pattern becomes. So this also indicates that this, you need a very specific frequency of light to see this effect. It needs a wavelength that is large enough compared to the object to see it.

In practice, you often have the combined effect. So if you have two slits, two slit experiment, but with the slit being multiples of wavelengths wide, so not infinitesimal slits, what you would typically do in a two-slit experiment, but slits of a finite width, then you would have, per slit, you would have a diffraction pattern, and then on top of that you would have an interference pattern caused by the two slits, and then you have this one modulating the interference pattern. So this is a single slit diffraction pattern, this is a two-slit interference pattern, and then the combined effect would be this envelope

convoluted with this interference pattern. So you would get an intensity pattern like this. So this is diffraction, the center, wide, right region, and on top of that you have the interference pattern due to the two slits. This gets complicated really quickly, so I'm happy already if you understand the individual concepts. This is just as an illustration, you would have the combined effects if you would have a real or a real life situation.

Yeah, I should really stop here. So we looked at dispersion and rainbows, many good discussions also, and that's why I run over time a bit. We looked at polarization and bruiser angle, scattering, interference, and diffraction. And before you leave, let me take five more minutes of your time, and then we have a 15-minute break.

Let me briefly do a man-to-meter quiz that also relates to the menti code.

Please enter this one, either by code, menti.com, use that code, or people are joining, good.

So how many do we have? Okay, a lot of people join, that's good. Someone still has to join or not?

I think most of you have joined. Okay, you can still join, the code will be at the top. So these are several questions also on the midterm. Did you attempt the midterm test?

I guess most of those who are here did attempt it, how many do we have?

68. I think I raised enough awareness for the midterm test. Okay, that's good.

If you don't want to answer, that's also fine.

What did you think of the feedback? Well, it's an optional midterm test, so there's really no loss if you just try it, and that's also how you could use it, so please do that. And it gives you an idea of what to expect at the final test.

Okay, how did you prepare for the midterm test? Let me see what you answered to this.

It's sometimes hard to break habits.

And if you just thought you knew the topics already and just wanted to see how you would do, that's also fine. That's why it's an optional test.

Yeah, this we see often. No time to prepare, that also happens.

Okay, thanks for that. I appreciate your feedback. There's a few questions now on attendance and resources. So this is probably some of you know a student body who inventorizes what's going on with courses during the quarter, during that they are running. This is also in cooperation with them that we try to investigate what resources you use for the course and how do you study, how do you prepare. I think you can select more than one here.

So what resources do you use for next two years?

How many people responded? Okay, I was expecting this, but not this much.

No, no, it's yeah.

Okay, if you don't attend, is that then caused by a clashing deadline sometimes or? There's three courses scheduled in principle, right? So you have computer

architecture, electronic circuits, one and this course.

But yeah, there might be deadlines for all of these. I know there was a midterm test last Monday for EC1. So that did not clash with physics, but may have affected your time to prepare. It may have caused less people to be here in the morning. I think there's someone holding down the thumbs up button.

Okay, two more slides, I think. Yeah, three statements. Let's see how much you agree with these. When would you skip a like live lecture? If there's recordings, does commute time to university affect your lecture, your attendance? Do the lectures provide added value compared to self-study?

I hope so, because I think I'm preaching to the choir here, though you are here, so you invest in coming through lectures, but still I want to get some insight on what happens.

Okay, let me end with two open questions.

Thank you so far. If there's one thing you would change that would make you attend lectures more, what would it be? So this is open questions. Is there anything that would make you attend lectures more? What would be it? And the next question will be the same, but for instructions. Yeah, fair point.

Really, you like Atlas 0820? I hate the room. I mean, there's one blackboard, two screens. Just in one morning, I get the mileage for a week or any. I do like walking around, but it's logistics-wise, I don't like it.

Yeah, but that's fair. And if you don't want to respond, that's also fine here. Faster-paced lectures, I need to strike a balance.

Power up. Going to bed earlier, yeah, that would help. It would help me as well.

Final question, and then we take the break.

More exam training. I will cover the midterm test in detail next studio classroom, so next Monday.

So last question, same question basically for instructions. So there might be different answers here. These are on Thursday afternoons. They may not be to your liking, or you may not like instructions in general. The instructions that are offered as separate things.

So on Thursday afternoon, those instructions, that's what I'm talking about here. So I scheduled one room only because of, well, experience with previous years. More rooms, yeah.

Over the past years, we started with four or five rooms, and after two weeks, we could go back to two and then to one room. It's a bit of a question and demand thing, but any feedback is welcome. It's anonymous, so I I will not judge you by it. I hope by now.